A

BRAC UNIVERSITY

Department of Computer Science & Engineering

Quiz 1

CSE 350: Digital Electronics and Pulse Techniques

Total Marks: 30 | TSE, Spring 26. | Time: 30 Minutes

Name:	ID:	Section:

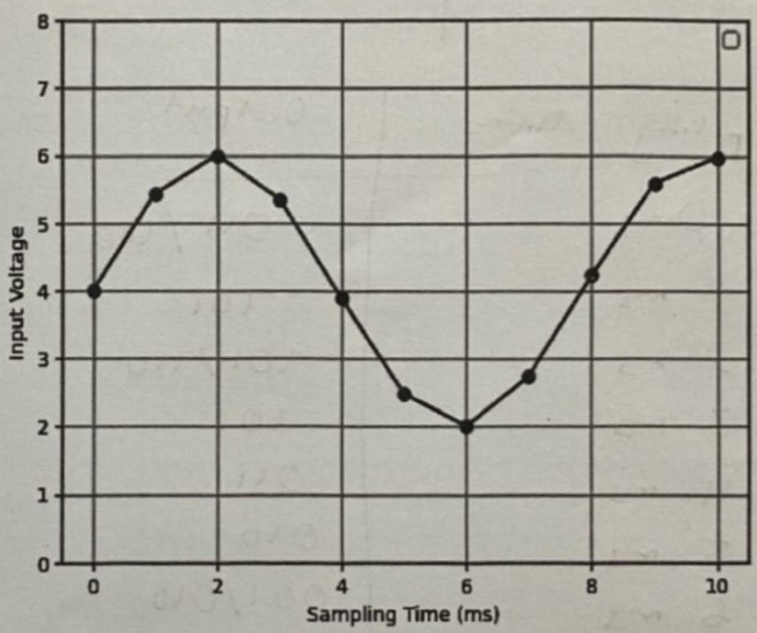
1. For a 3 bit Flash ADC with $V_{max} = 8V$ and $V_{min} = 0V$. Answer the following questions :

   (a) Calculate the total number of resistors and Op amps required [3 mark]

   (b) Calculate the step size [4 mark]

   (c) Find the **quantization range, quantization level** and respective *Digital Output* using **mid rise** quantization scheme. [4 mark]

   (d) For the given sampled signal, find the encoded digital output. [3 mark]



   (e) If the digital output is $(110)_2$, what are the maximum and minimum values of input that would produce this result? [2 mark]

   (f) Draw the Flash ADC. [4 mark]

a) Resistors - 8 , Op amp - 7

b) Step size $= \frac{8-0}{2^3} = 1V$

c)

Quantization range	Quantization level	Digital output
0 - 1	0.5V	000
1 - 2	1.5V	001
2 - 3	2.5V	010
3 - 4	3.5V	011
4 - 5	4.5V	100
5 - 6	5.5V	101
6 - 7	6.5V	110
7 - 8	7.5V	111

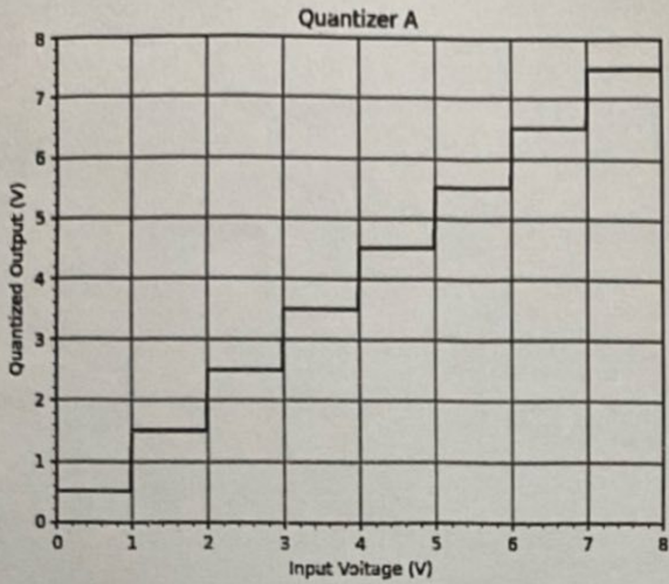
d)

Sampling times	Output
0 ms	011 / 100
1 ms	101
2 ms	101 / 110
3 ms	101
4 ms	011
5 ms	010
6 ms	001 / 010
7 ms	010
8 ms	100
9 ms	101
10 ms	101

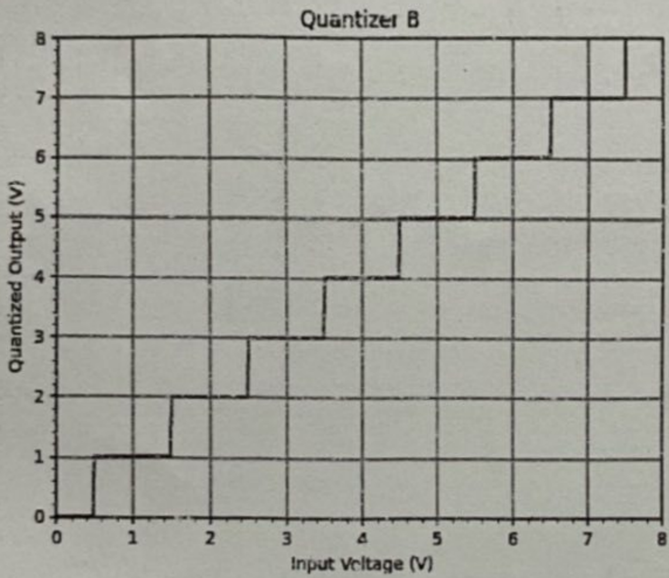
e) 6V - 7V

f) Draw ADC with 8 resistors and 7 op amps.

2. For the given Quantization schemes:



Mid rise



Mid tread

(a) Identify the quantization schemes used in both the quantizers. [2 mark]

(b) For both the quantizers, find the quantized output and the digital output if the input voltage is 5.2 V [8 mark]

b) Mid rise

Quantized output - 5.5V, Digital output - 101

Midtread

Quantized output - 5V, Digital - 101